

THE CYCLIC ANTIBANDWIDTH OF THE $3 \times 3 \times 4$ MESH IS 13

ABSTRACT. For a graph G on n vertices, a labeling is a bijection $f: V(G) \rightarrow \{1, \dots, n\}$, and the cyclic antibandwidth $\text{cab}(G)$ is the largest k for which some labeling places the two ends of every edge at distance at least k apart on the cycle $\mathbb{Z}_n$. We prove $\text{cab}(P_3 \times P_3 \times P_4) = 13$. The lower bound is a printed 36-label bijection, certified by 75 integer comparisons. The upper bound is a complete backtracking exhaustion at $k = 14$, using no solver and no library, run twice — once under the symmetry reduction argued here and once with no symmetry reduction at all; a degree count disposes of $k \geq 16$ by hand. The appendix table of conjectured 3-dimensional-mesh values reproduced in the 2026 exact-method paper of Xuan and Van lists 14 at this instance, one above the value proved here. What that table records, and by whom the entry was posed, are not settled here; nothing is claimed about any other instance.

1. INTRODUCTION

Let G be a graph on $n = |V(G)|$ vertices. A *labeling* of G is a bijection $f: V(G) \rightarrow \{1, \dots, n\}$. For labels a, b put

$$d_c(a, b) = \min(|a - b|, n - |a - b|),$$

the distance between a and b on the cycle $\mathbb{Z}_n$, and set

$$\text{cab}_f(G) = \min_{uv \in E(G)} d_c(f(u), f(v)), \quad \text{cab}(G) = \max_f \text{cab}_f(G).$$

This is the *cyclic antibandwidth* of G . It is a max-min parameter, so a single labeling certifies a lower bound and nothing else: one edge sets the value of cab_f , and a claimed witness is checked in $O(|E|)$ integer arithmetic with no search. The 3-dimensional mesh is the Cartesian product $P_{n_1} \times P_{n_2} \times P_{n_3}$ of paths.

Exact values of cab are known for paths, cycles, two-dimensional and toroidal meshes and hypercubes [3, 2], as surveyed in [5]; for three-dimensional meshes what the literature offers is a table of conjectured values obtained from a heuristic. Nothing below uses the contents of [3, 2, 4].

One such table is the source of the value we compare with. The exact-method paper of Xuan and Van [5] reproduces, in its appendix on the experimental datasets, a 3-dimensional mesh table captioned “Conjectured

2020 *Mathematics Subject Classification*. 05C78, 05C85, 68R10.

Key words and phrases. cyclic antibandwidth, antibandwidth, three-dimensional mesh, graph labeling, exact value, exhaustive search.

CAB value of 3D mesh instances. Source: Bansal et al.”; in the row for $3 \times 3 \times n_3$ meshes, read from the e-print [arXiv:2601.04239v1](#) (source lines 1398–1422), the entry at $n_3 = 4$ is 14. In [5] the tabulated values are used to set the search bound for 3-dimensional meshes, which “is chosen as the value immediately above the conjectured CAB value reported by Bansal et al.” (source line 1395), with the caveat that “since this value is conjectured, it may be less than the actual optimal CAB value” (line 1534).

Theorem 1. $\text{cab}(P_3 \times P_3 \times P_4) = 13$.

So the tabulated entry 14 is one above the cyclic antibandwidth of that instance, and it is unattained: no labeling of $P_3 \times P_3 \times P_4$ reaches 14.

Remark. We do not claim to have refuted a stated conjecture. We did not read the body of [1], which is closed access, so we cannot say what closed form, if any, its authors posed; the accessible source is the dataset caption quoted above. Nor do we establish which parameter the entry records. The recorded run accompanying this note reports that the row’s entries at the smallest six values of n_3 coincide with values reported for the *linear* antibandwidth of the same meshes, so the printed 14 may be a linear rather than a cyclic value; on that reading it is not in conflict with anything proved here. What is proved here is the displayed theorem.

Section 2 exhibits a witness for $\text{cab} \geq 13$; six of its 75 comparisons are displayed and the rest are the same integer arithmetic. Section 3 bounds cab above: a degree count gives $\text{cab} \leq 15$ by hand, and a complete backtracking exhaustion at $k = 14$ of a reformulation (Lemma 3) gives $\text{cab} \leq 13$, using no solver, no library and no floating-point arithmetic. Section 4 says what is not settled.

2. THE LOWER BOUND: A PRINTED WITNESS

Write a vertex of $P_3 \times P_3 \times P_4$ as (x, y, z) with $x, y \in \{1, 2, 3\}$ and $z \in \{1, 2, 3, 4\}$; two vertices are adjacent when they differ by 1 in exactly one coordinate. So $n = 36$ and $|E| = 75$. The labeling W below is printed as four rows $z = 1, \dots, 4$; each row groups $x = 1 \mid x = 2 \mid x = 3$ and within a group lists $y = 1, 2, 3$.

$W:$	$z = 1$	31	14	30	13	29	10	35	12	27
	$z = 2$	15	1	17	36	16	32	21	34	11
	$z = 3$	2	23	4	22	3	18	7	20	33
	$z = 4$	24	8	26	9	25	5	28	6	19

Proposition 2. W is a labeling of $P_3 \times P_3 \times P_4$ with $\text{cab}_W = 13$. Hence $\text{cab}(P_3 \times P_3 \times P_4) \geq 13$.

Proof. The table lists each of $1, \dots, 36$ exactly once, so W is a bijection. For each of the 75 edges one computes d_c with $n = 36$; the minimum is 13, attained on 15 edges. Nothing but integer arithmetic is involved. As an

illustration, take the degree-6 vertex $(2, 2, 2)$, which carries label 16; its six neighbours

$$(1, 2, 2), (3, 2, 2), (2, 1, 2), (2, 3, 2), (2, 2, 1), (2, 2, 3)$$

carry 1, 34, 36, 32, 29, 3 respectively, and

$$d_c(16, \cdot) = 15, 18, 16, 16, 13, 13,$$

all at least 13. $\square$

3. THE UPPER BOUND

Everything in this section rests on one elementary reformulation. For a label $l \in \mathbb{Z}_n$ put

$$F_k(l) = \{l' \in \mathbb{Z}_n : d_c(l, l') \geq k\} = \{l + k, l + k + 1, \dots, l + n - k\},$$

an arc of exactly $n - 2k + 1$ labels antipodal to l .

Lemma 3. *Let G have n vertices and let $1 \leq k \leq \lfloor n/2 \rfloor$. For a labeling f the following are equivalent.*

- (i) $\text{cab}_f(G) \geq k$.
- (ii) *For every edge uv , $f(v) \in F_k(f(u))$; equivalently, G is a spanning subgraph of the circulant $\text{Ci}_n(k, k+1, \dots, \lfloor n/2 \rfloor)$ under f .*
- (iii) *For every $s \in \mathbb{Z}_n$ the set $f^{-1}(\{s, s+1, \dots, s+k-1\})$ of vertices carrying k cyclically consecutive labels is independent in G .*

Proof. (i) $\Leftrightarrow$ (ii) is the definition of F_k , and the circulant with connection set $\{k, \dots, \lfloor n/2 \rfloor\}$ is precisely the graph on $\mathbb{Z}_n$ whose edges join labels at cyclic distance at least k . For (i) $\Leftrightarrow$ (iii): two labels in a common window of k cyclically consecutive labels are at cyclic distance at most $k-1$, and conversely two labels at cyclic distance $d \leq k-1$ lie in a common such window. So some window carries an edge iff some edge has $d_c < k$. $\square$

Two consequences.

Lemma 4 ($k \geq 16$). $\text{cab}(P_3 \times P_3 \times P_4) \leq 15$.

Proof. Let $\Delta = \Delta(G)$ and let v have degree Δ . By Lemma 3(ii) the Δ neighbours of v receive distinct labels from the $n - 2k + 1$ labels of $F_k(f(v))$, so $\text{cab}(G) \leq \lfloor (n - \Delta + 1)/2 \rfloor$. Here $n = 36$ and $\Delta = 6$, attained exactly at $(2, 2, 2)$ and $(2, 2, 3)$, so $\text{cab} \leq \lfloor 31/2 \rfloor = 15$. Concretely, $k = 16$ leaves only $36 - 31 = 5$ admissible labels for six neighbours. $\square$

Neither of the two counting obstructions available here excludes $k = 14$: the host circulant $\text{Ci}_{36}(14, \dots, 18)$ is 9-regular, which is at least $\Delta = 6$, and by Lemma 3(iii) a window of 14 labels must be independent, which is no obstruction either, since $P_3 \times P_3 \times P_4$ has a perfect matching and hence independence number $36 - 18 = 18 > 14$. We decide $k = 14$ by exhaustion.

Lemma 5 ($k = 14$). *There is no labeling f of $P_3 \times P_3 \times P_4$ with $\text{cab}_f \geq 14$.*

Proof (a complete backtracking exhaustion). By Lemma 3(ii) with $k = 14$ we must decide whether $G = P_3 \times P_3 \times P_4$ embeds as a spanning subgraph of $\text{Ci}_{36}(14, \dots, 18)$. Order the 36 vertices $v_1, \dots, v_{36}$ deterministically: v_1 is the vertex of largest degree that is least in the lexicographic order of (x, y, z) , so $v_1 = (2, 2, 2)$; and v_{i+1} is the unordered vertex with the most already-ordered neighbours, ties broken by larger degree and then by lexicographic order. Assign labels to $v_1, v_2, \dots$ in that order by depth-first search, trying admissible labels in increasing order: a label is admissible for v_i if it is unused and lies in $F_{14}(f(w))$ for every already-labelled neighbour w of v_i . After each assignment, propagate — intersect the candidate arc of every not-yet-labelled neighbour, and backtrack as soon as one of them is left with no admissible label. This decides the question exactly: the search enumerates every labeling with the required property, and it terminates.

Two reductions cut the tree, and both are exact. *Rotation.* $f \mapsto f + t$ preserves every cyclic distance and acts transitively on which vertex carries a given label, so we may fix $f(v_1) = 0$ (labels are taken in $\mathbb{Z}_{36}$ here). *The automorphisms of the layer.* The eight automorphisms of the 3×3 grid fix $v_1 = (2, 2, 2)$ and act transitively on its four in-layer neighbours, so we may require the least of their four labels to sit on $(1, 2, 2)$; the one non-identity grid automorphism fixing $(1, 2, 2)$ transposes $(2, 1, 2)$ and $(2, 3, 2)$, so we may require $f(2, 1, 2) < f(2, 3, 2)$ as well. Finally $f \mapsto -f$ fixes the label 0, reverses every comparison and commutes with the grid automorphisms, which do not touch z , so we may require $f(2, 2, 1) < f(2, 2, 3)$. Together these quotient a group of order 16 acting on the solution set.

Executed at $k = 14$, the search reports no labeling in either run:

reduced,	281,957,255 nodes,	0 labelings;
no symmetry reduction,	2,463,942,873 nodes,	0 labelings.

In the reduced run the tree never reaches beyond its 31st level: every branch is abandoned with at least five vertices still unlabelled. The second run uses no symmetry reduction at all beyond the rotation pin, so it does not depend on the argument of the previous paragraph, and it agrees. $\square$

Proof of Theorem 1. $\text{cab} \geq 13$ by Proposition 2, and $\text{cab} \leq 13$ by Lemma 5. $\square$

4. STATUS

What is settled is $\text{cab}(P_3 \times P_3 \times P_4) = 13$. Every layer of that proof is re-derived by the programs accompanying this note, from the labeling printed in Section 2, with no solver and no third-party library.

What is not settled:

- *No other instance.* Nothing here bounds $\text{cab}(P_3 \times P_3 \times P_{n_3})$ for any $n_3 \neq 4$, and the tabulated row is not repaired.

- *The status of the tabulated 14.* Who posed it, in what form, and which parameter it records are all outside this note; see the remark in Section 1.
- *The $k = 14$ layer is a computation, not a readable proof.* It is solver-free, exhaustive and deterministic, but it is a search of hundreds of millions of nodes: a referee must re-run it rather than read it, and no independently checkable refutation object (a DRAT or LRAT trace, or a hand-sized infeasible core) accompanies it.

5. REPRODUCTION

Everything up to and including Lemma 4 needs no code: rebuild $P_3 \times P_3 \times P_4$, confirm $|V| = 36$, $|E| = 75$ and $\Delta = 6$ at exactly $(2, 2, 2)$ and $(2, 2, 3)$; confirm that the table of W is a bijection onto $\{1, \dots, 36\}$; compute the 75 cyclic distances and read off the minimum 13; and check the arithmetic of Lemma 4. That is one page, and it gives $13 \leq \text{cab} \leq 15$.

The exhaustion of Lemma 5 is machine work. Two programs accompany this note, each with a recorded run: `verify.py` with `verify.output.txt` re-derives the objects and the reduced $k = 14$ exhaustion, and `exhaust.py` with `exhaust.output.txt` re-decides $k = 14$ with the symmetry reduction removed entirely. Both need Python 3.9 or later and the standard library only — no third-party package, no external data file, no network, no randomness and no floating-point value in any decision — and both are deterministic, so the node counts printed above are reproducible exactly. Run them as `python3 verify.py` and `python3 exhaust.py`; in the recorded runs they took about three minutes and about twenty-three minutes on one core. Each program closes with a list of what it does not do; neither checks a bibliographic locator, and neither bears on prior art.

REFERENCES

- [1] R. Bansal and K. Srivastava, *A memetic algorithm for the cyclic antibandwidth maximization problem*, Soft Comput. **15** (2011), no. 2, 397–412. doi:10.1007/s00500-009-0538-6. Cited by [5] as the source of the tabulated values; the body is closed access and we did not read it.
- [2] A. Raspaud, H. Schröder, O. Sýkora, L. Török, and I. Vrto, *Antibandwidth and cyclic antibandwidth of meshes and hypercubes*, Discrete Math. **309** (2009), no. 11, 3541–3552. doi:10.1016/j.disc.2007.12.058.
- [3] O. Sýkora, L. Török, and I. Vrto, *The cyclic antibandwidth problem*, Electron. Notes Discrete Math. **22** (2005), 223–227. doi:10.1016/j.endm.2005.06.030.
- [4] L. Török and I. Vrto, *Antibandwidth of three-dimensional meshes*, Discrete Math. **310** (2010), no. 3, 505–510. doi:10.1016/j.disc.2009.03.029.
- [5] H. T. Xuan and K. T. Van, *Solving cyclic antibandwidth problem by SAT*, Comput. Optim. Appl. (2026), doi:10.1007/s10589-026-00819-8; e-print arXiv:2601.04239v1. The caption and the entry quoted in Section 1 are read from the e-print source, in the appendix table of conjectured 3D-mesh CAB values, lines 1398–1422.